



Enderass National PLC Digital Auction System

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS)

1. INTRODUCTION

1.1 Purpose

The purpose of this system is to digitize and automate the auction management process of Enderass National PLC. The platform will facilitate communication and transactions between auction participants and Enderass operational staff through a secure online auction management system.

The system will support asset evaluation, auction publication, user registration, document purchasing, bid submission, winner selection, notification management, reporting, and internal operational processes.

A single registered user may participate in auctions as a bidder and may also submit their own assets to be auctioned through Enderass. The system shall not require separate accounts for bidders and asset owners.

Language Support:

- English
- Amharic

Platforms:

- Web Application
- Android Mobile Application
- iOS Mobile Application

1.2 Scope

The system shall provide:

- Public auction portal
- User management
- Asset owner management
- Bidder management
- Auction management
- Asset evaluation management
- Bid submission management
- CPO verification management
- Payment management
- Notification management
- Reporting and analytics
- Staff management
- Role-based access control

The platform shall support closed bidding processes according to Ethiopian auction practices.

2. USER TYPES

2.1 Registered User (Bidder and Asset Owner)

Individuals or organizations using the platform.

A registered user may:

- Participate in auctions as a bidder
- Submit their own assets for auction
- Track submitted assets
- Track auction participation
- Receive notifications related to both activities

Capabilities:

- Register account
- Login via OTP
- Browse auctions
- Purchase auction documents
- Submit CPO
- Participate in auctions
- Submit auction requests for owned assets
- Upload asset details and supporting documents
- Track asset evaluation and auction status
- Track participation status
- Receive notifications
- View auction results
- Manage profile information

Examples:

- Individual property owners
- Insurance companies
- Banks
- Government organizations
- NGOs
- Private companies
- Businesses participating in auctions

2.2 Enderass Operational Users

Internal users responsible for managing operations.

Capabilities:

- Manage auction requests
- Evaluate assets
- Create auctions
- Verify payments
- Verify CPO submissions
- Manage users
- Generate reports
- Send notifications

2.3 Super Administrator

System owner with full privileges.

Capabilities:

- View all data
- Manage roles
- Open encrypted bids
- Select winners
- Configure system settings
- Audit system activities

3. BUSINESS PROCESS

Phase 1: Auction Request

1. Registered user submits an auction request for their asset.
2. Request enters pending review.
3. Enderass reviews the request.
4. Asset evaluation is scheduled.

Phase 2: Asset Evaluation

1. Evaluation officer inspects the asset.
2. Valuation report is prepared.
3. Supporting photos are uploaded.
4. Asset approved or rejected.

Phase 3: Auction Creation

Operational user creates auction:

- Title

- Category
- Description
- Images
- Documents
- Start date
- Closing date
- CPO percentage
- Document fee
- Reserve price
- Auction conditions

Phase 4: Auction Participation

1. User purchases auction document.
2. User downloads document.
3. User uploads CPO evidence.
4. CPO verified by Enderass.
5. User submits sealed bid.

Phase 5: Auction Closure

1. Auction automatically closes.
2. Bid submissions locked.
3. Winner selection process begins.

Phase 6: Winner Declaration

1. Authorized personnel open bids.
2. Winner selected.
3. Notifications sent.
4. Auction marked completed.

4. FUNCTIONAL REQUIREMENTS

4.1 Authentication Module

Features:

- SMS OTP Registration
- SMS OTP Login
- Passwordless Authentication
- Session Management
- Device Tracking

Data:

- Full Name

- Company Name (Optional)
- Phone Number

4.2 Auction Management Module

Features:

- Create Auction
- Edit Auction
- Publish Auction
- Suspend Auction
- Close Auction
- Archive Auction

Auction Categories:

- Vehicles
- Machinery
- Buildings
- Land
- Equipment
- Salvage Assets
- Other Assets

4.3 Asset Evaluation Module

Features:

- Asset Inspection
- Valuation Recording
- Evaluation Reports
- Approval Workflow
- Asset History Tracking

4.4 Auction Request Module

Features:

- Submit Asset for Auction
- Upload Asset Information
- Upload Photos and Documents
- Track Evaluation Status
- Track Auction Creation Status
- View Auction Outcome
- Communication with Enderass

A registered user shall be able to submit one or more assets for auction through both the web application and mobile application.

4.5 Auction Document Management

Features:

- Upload Documents
- PDF Preview
- Paid Access Control
- Watermarked Downloads
- Version Control

4.6 Payment Module

Supported Payment Methods:

- Telebirr
- CBE Birr
- Awash Wallet
- Addis Pay/SantimPay/chapa

Functions:

- Document Purchase
- Payment Verification
- Refund Processing
- Receipt Generation

4.7 CPO Management Module

Features:

- Upload CPO
- Verification Workflow
- Approval/Rejection
- Expiration Tracking
- Audit History

System shall automatically calculate required CPO amount based on auction settings.

4.8 Closed Bid Module

Features:

- Sealed Bid Submission
- Bid Encryption
- Bid Locking

- Bid Validation
- Bid History

Requirements:

- No bidder can view another bidder's bid.
- Users can view only their own submitted bids.
- Operational staff cannot view submitted amounts.
- Only authorized super administrators can access bids after auction closure.

4.9 Winner Management Module

Features:

- Winner Selection
- Winner Notification
- Acceptance Tracking
- Alternate Winner Selection
- Award Documentation

4.10 Notification Module

Channels:

- SMS
- Email
- In-App Notifications

Events:

- Registration
- Payment Confirmation
- CPO Approval
- Auction Reminder
- Auction Closure
- Winner Announcement
- Asset Evaluation Updates
- Auction Request Status Updates

4.11 Reporting Module

Reports:

- Active Auctions
- Closed Auctions
- Revenue Reports
- Asset Owner Reports

- Bidder Reports
- Participation Reports
- CPO Reports
- Payment Reports

Export Formats:

- PDF
- Excel
- CSV

4.12 User Management Module

Features:

- User Registration
- User Verification
- User Suspension
- User Activity Logs
- Role Assignment

The system shall maintain a single user account that can perform both bidder and asset owner activities.

4.13 Audit Trail Module

System shall record:

- User actions
- Login history
- Bid submissions
- Asset submissions
- Payment activities
- Administrative actions

Audit logs cannot be modified.

5. ROLE-BASED ACCESS CONTROL

Registered User

Capabilities:

- Participate in auctions
- Purchase auction documents
- Submit CPO
- Submit bids

- View own bids
- Submit assets for auction
- Track asset evaluation
- Track auction requests
- View auction outcomes
- Receive notifications
- Super Administrator
 - Full system access.
- Auction Manager
 - Manage auctions and monitor operations.
- Evaluation Officer
 - Manage asset evaluations.
- Finance Officer
 - Manage payments and financial reports.
- Customer Service Officer
 - Manage customer support and communications.
- Auditor
 - Read-only access to reports and logs.

6. WEB AND MOBILE APPLICATION REQUIREMENTS

Unified User Application

The system shall provide a single web application and a single mobile application for all external users.

There shall be no separate bidder application or asset owner application.

A registered user shall be able to:

- Register and login
- Browse auctions
- Purchase auction documents
- Submit CPO
- Submit bids
- View bid history
- Track auction participation
- Submit assets for auction

- Upload asset details and documents
- Track asset evaluation progress
- Monitor auction status of submitted assets
- Receive notifications
- Manage profile information

The same account shall allow users to act as both bidders and asset owners simultaneously.

Mobile Application Features

Features:

- Registration and Login
- Auction Browsing
- Auction Search and Filtering
- Document Purchase
- CPO Submission
- Bid Submission
- Bid Tracking
- Asset Submission
- Asset Tracking
- Notifications
- Profile Management

Web Application Features

Features:

- Registration and Login
- Auction Browsing
- Auction Search and Filtering
- Document Purchase
- CPO Submission
- Bid Submission
- Bid Tracking
- Asset Submission
- Asset Tracking
- Notifications
- Profile Management

7. NON-FUNCTIONAL REQUIREMENTS

Performance

- Support 10,000+ users
- Support 500+ concurrent users
- Page load under 3 seconds

Security

- OTP Authentication
- Data Encryption
- SSL Security
- Role-Based Access
- Audit Logging
- Secure File Storage

Availability

- 99.5% uptime
- Daily backups
- Disaster recovery procedures

Scalability

System must support future expansion including:

- Multi-branch operations
- Multiple auction companies
- Additional payment gateways
- Regional language support

8. FUTURE FEATURES

- Online Live Auction/Open Bid
- AI Asset Valuation Assistance
- Digital Contract Signing
- National Auction Marketplace
- API Integration with Banks
- Government Asset Auction Portal

9. SUCCESS METRICS

The system shall achieve:

- Reduced manual paperwork
- Faster auction processing

- Increased bidder participation
- Improved transparency
- Improved operational efficiency
- Accurate reporting and compliance
- Secure closed bidding process

Mager Software PLC.